Arkil Patel

PhD Student, McGill University *and* **Mila**

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D-2107N, Mila - Quebec AI Institute, 6666 Rue Saint-Urbain, Montréal, QC H2S 3H1

Education

McGill University, *Montreal, Canada* 08/2022 - Present
Ph.D. in Computer Science (ongoing), GPA: 4.0/4.0
Advisors: Prof. Dzmitry Bahdanau and Prof. Siva Reddy

Birla Insitute of Technology and Science, *Goa, India* 08/2016 - 07/2020
B.E. in Computer Science
Graduated with *distinction*, GPA: 9.03/10

Experience

Mila - Quebec AI Institute, *Montreal, Canada* 08/2022 - Present
Research Assistant
Advisors: Prof. Dzmitry Bahdanau and Prof. Siva Reddy
› Working on scaling laws, forecasting, generalization, safety, and evaluation.

Allen Institute for Artificial Intelligence (AI2), *Seattle, USA* 05/2023 - 08/2023
Research Intern, AllenNLP
Advisor: Dr. Pradeep Dasigi
› Worked on evaluating the code generation abilities of LLMs.

Microsoft Research, *Bangalore, India* 09/2020 - 07/2022
Pre-doctoral Research Fellow
Advisor: Dr. Navin Goyal
› Developed models that generalize compositionally on semantic parsing and grounded language understanding tasks.

Microsoft Research, *Bangalore, India* 06/2019 - 12/2019
Research Intern
Advisor: Dr. Navin Goyal
› Developed robust and interpretable models for semantic parsing.

Publications

P=Preprint, C=Conference, J=Journal, W=Workshop [Citations: 3000+, h-index: 14, i10-index: 15]

C.12 **Forecasting Downstream Performance of LLMs With Proxy Metrics**
Arkil Patel, Siva Reddy, Marius Mosbach, Dzmitry Bahdanau
Accepted to NeurIPS 2026 **[NeurIPS 2026]**

P.2 **Toward Open Weight Models Without Risks: Separating Public and Private Capabilities in LLMs**
Arkil Patel*, Charbel El Feghali*, Nicholas Meade, Spandana Gella, Verna Dankers, Siva Reddy
In submission to ICLR 2027

P.1 **How to Get Your LLM to Generate Challenging Problems for Evaluation**
Arkil Patel, Siva Reddy, Dzmitry Bahdanau
In submission

W.1 **Understanding Scaling Laws With Token-Level Analysis**
Arkil Patel, Marius Mosbach, Siva Reddy, Dzmitry Bahdanau
Workshop on Scientific Methods for Understanding Deep Learning **[Sci4DL 2026]**

J.3 **DeepSeek-R1 Thoughtology: Let's think about LLM Reasoning**
Arkil Patel*, Sara Vera Marjanović*, et al. (* = Equal Contribution)
Transactions of Machine Learning Research **[TMLR 2026]**

C.11 **Agentrewardbench: Evaluating automatic evaluations of web agent trajectories**
Xing Han Lù, Amirhossein Kazemnejad, Nicholas Meade, Arkil Patel, Dongchan Shin, Alejandra Zambrano, Karolina Stańczak, Peter Shaw, Christopher J. Pal, Siva Reddy
Conference on Language Modeling **[COLM 2025]**

J.2 **Investigating Adversarial Trigger Transfer in Large Language Models**
Nicholas Meade, Arkil Patel, Siva Reddy
Transactions of the Association for Computational Linguistics **[TACL 2025]**

C.10 **Safearena: Evaluating the safety of autonomous web agents**
Ada Defne Tur, Nicholas Meade, Xing Han Lù, Alejandra Zambrano, Arkil Patel, Esin Durmus, Spandana Gella, Karolina Stańczak, Siva Reddy
Forty-Second International Conference on Machine Learning **[ICML 2025]**

J.1 **Humanity's last exam**
Long Phan, ..., Arkil Patel, et al.
Nature **[Nature 2026]**

C.9 **Evaluating In-Context Learning of Libraries for Code Generation**
Arkil Patel, Siva Reddy, Dzmitry Bahdanau, Pradeep Dasigi
2024 Conference of North American Chapter of the Association for Computational Linguistics **[NAACL 2024]**

C.8 **Understanding In-Context Learning in Transformers and LLMs by Learning to Learn Discrete Functions**
Satwik Bhattamishra, Arkil Patel, Phil Blunsom, Varun Kanade
The Twelfth International Conference on Learning Representations **[ICLR 2024]**

C.7 **MAGNIFICo: Evaluating the In-Context Learning Ability of Large Language Models to Generalize to Novel Interpretations**
Arkil Patel, Satwik Bhattamishra, Siva Reddy, Dzmitry Bahdanau
2023 Conference on Empirical Methods in Natural Language Processing **[EMNLP 2023]**

C.6 **Simplicity Bias in Transformers and their Ability to Learn Sparse Boolean Functions**
Satwik Bhattamishra, Arkil Patel, Varun Kanade, Phil Blunsom
61st Annual Meeting of the Association for Computational Linguistics **[ACL 2023]**

C.5 **When Can Transformers Ground & Compose: Insights from Compositional Generalization Benchmarks**
Ankur Sikarwar, Arkil Patel, Navin Goyal
2022 Conference on Empirical Methods in Natural Language Processing **[EMNLP 2022]**

C.4 **Revisiting the Compositional Generalization Abilities of Neural Sequence Models**
Arkil Patel, Satwik Bhattamishra, Phil Blunsom, Navin Goyal
60th Annual Meeting of the Association for Computational Linguistics **[ACL 2022]**

C.3 **Are NLP Models really able to Solve Simple Math Word Problems?**
Arkil Patel, Satwik Bhattamishra, Navin Goyal
2021 Conference of North American Chapter of the Association for Computational Linguistics **[NAACL 2021]**

C.2 **On the Computational Power of Transformers and Its Implications in Sequence Modeling**
Satwik Bhattamishra, Arkil Patel, Navin Goyal
2020 Conference on Computational Natural Language Learning **[CoNLL 2020]**

C.1 **VehicleChain: Blockchain-based Vehicular Data Transmission Scheme for Smart City**
Arkil Patel, Naigam Shah, Trupil Limbasiya, Debasis Das
2019 IEEE International Conference on Systems, Man and Cybernetics **[SMC 2019]**

Relevant Research

Forecasting Capabilities of Large Language Models with Scaling Laws [ ] 03/2025 - Present
w/ Dzmitry Bahdanau, Siva Reddy, Marius Mosbach
› Developing general hypotheses for scaling laws for downstream tasks.
› Designing proxy evaluation measures for predicting LLM capabilities that generalize for new tasks and unseen LLMs.
Under submission to NeurIPS 2026

Understanding Power Law Scaling in LLMs [ ] 03/2025 - Present
w/ Dzmitry Bahdanau, Siva Reddy, Marius Mosbach
› Analysed individual token-level scaling behaviour in LLM pretraining.
› Found that while individual token-level trajectories are noisy, most of them are still well-approximated by power laws.

Synthetic Data Generation for Evaluation [ ] 12/2023 - 10/2024
w/ Dzmitry Bahdanau, Siva Reddy
› Developed a general framework to scalably generate challenging problems from LLMs for evaluation.
› Created automatic benchmarks for document-based QA, repository-level code completion, and math reasoning.
› Experiments over 15 contemporary LLMs showed that all models struggled on our synthetic benchmarks.
In submission

Evaluating Large Language Models: Acquisition and Usage of Code Modules [ ] 05/2023 - 11/2023
w/ Pradeep Dasigi, Dzmitry Bahdanau, Siva Reddy
- Developed a framework to assess the ability of LLMs to in-context learn novel libraries and code functions.
- Showed that even smaller, open-source models such as StarCoder can learn code libraries in-context and that models such as GPT-4 can learn as efficiently from natural language descriptions as they can from demonstrations.

Presented at NAACL 2024

Compositional Generalization in Large Language Models [ ] 01/2023 - 06/2023
w/ Dzmitry Bahdanau, Siva Reddy
- Developed a benchmark based on text-to-SQL to assess the ability of LLMs to generalize to novel concepts.
- Highlighted various capabilities and limitations associated with in-context compositional generalization in LLMs.

Presented at EMNLP 2023 [Oral]

Awards and Honors

Canada Graduate Scholarships (Doctoral) [ ] *Awarded funding of $40,000/yr for 3 years* Montreal, 05/2025
Graduate Excellence Award [ ] *Awarded funding of $6,000* Montreal, 09/2024
Canada Graduate Scholarships (Masters) [ ] *Awarded funding of $17,500* Montreal, 05/2023
Institute Merit Scholarship [ ] *Tuition fee waiver for ranking in top 1% of all students* Goa, 08/2016 - 07/2020
National Talent Search Scholarship [ ] *Awarded to top 1000 students in the country.* New Delhi, 08/2012 - 07/2020

Teaching and Leadership Roles

Large Language Models, COMP 767 @ McGill. *Teaching Assistant w/ Siva Reddy* Winter 2026
Natural Language to Data Science, COMP 345 @ McGill. *Teaching Assistant w/ Siva Reddy* Winter 2025
Large Language Models, COMP 767 @ McGill. *Teaching Assistant w/ Siva Reddy* Fall 2024
Natural Language to Data Science, COMP 345 @ McGill. *Teaching Assistant w/ Siva Reddy* Winter 2024
Natural Language to Data Science, COMP 345 @ McGill. *Teaching Assistant w/ Siva Reddy* Winter 2023
Neural Networks, BITS F312 @ BITS-Pilani. *Teaching Assistant w/ Basabdatta Sen Bhattacharya* Winter 2020
Data Mining, CS F415 @ BITS-Pilani. *Teaching Assistant w/ Hemant Rathore* Winter 2019
Database Systems, CS F212 @ BITS-Pilani. *Teaching Assistant w/ Debasis Das* Winter 2019
Speech & NLP Reading Group, Microsoft Research India. *Organizer* 2021 - 2022

Invited Talks

HEC MATH 60621A NLP Guest Lecture on *"Transformer-based Encoders"* Montreal, 03/2026
Invited Talk at BITS Pilani on *"Evaluation and Forecasting in the Era of LLMs"* Remote, 11/2025
McGill COMP 599 NLU w/ Deep Learning Guest Lecture on *"In-Context Learning"* Montreal, 02/2024
McGill COMP 345 Data Science Lecture on *"Decoding from Language Models and Beam Search"* Montreal, 03/2023
Microsoft Research Invited Talk on *"Compositional Generalization in Language Models"* Bangalore, 03/2022

Academic Service

Reviewer NeurIPS'26, ICLR'25, ICML'25, NeurIPS'25, COLM'24, ARR'23, EMNLP'23, ACL'23, ARR'22, EMNLP'22, AAAI'22
Sub-Reviewer NAACL'21, EMNLP'21
Volunteer EMNLP'22, Panini Linguistics Olympiad (PLO) '21